hand-drawn, presented live

Flowcam

Business model and cost report

A plain-language guide to the opportunity, the product, how Flowcam makes money and what it costs to run. No prior knowledge of the product or of software businesses is needed.

76%
of each $17 subscription left after per-customer costs

7
paying customers cover the monthly running costs

$52
fixed running costs per month at launch

~$700
starting budget, covering the build and four months after launch

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25 October 2026 · Status: pre-launch, no paying customers yet.
Based on Flowcam's product spec and decision records (17 October 2026), its open-items list and its build history. Every figure here is a plan or an estimate, not a result.

AT A GLANCE

Flowcam in one page

Flowcam is a studio for making explainer videos. You draw a diagram in a hand-drawn style, then present it step by step while your face, voice and screen are recorded. Flowcam then helps you polish the result into a professional video.

Question	Short answer
Who is it for?	People who explain things on video: developer advocates and engineers explaining systems, teachers and course creators, and explainer YouTubers. Individuals first; teams later. Sold worldwide, in English, priced in US dollars.
What makes it different?	No other product combines a step-by-step diagram you design in advance with a synchronised face camera. Flowcam also stores a recording as instructions rather than as frozen video, so you can fix a typo after recording, re-export in any size, and let viewers copy the diagram.
How does it make money?	A free version that works on your own computer, and a Pro subscription at $20 a month ($17 for early "Founding" subscribers) for online features. AI features are paid for with Credits: Pro includes 1,000 a month, and extra Credits can be bought.
What does a customer cost?	About $4 of every $17 goes to services and payment fees. About $13 is left.
When does it break even?	About 7 paying customers cover the monthly running costs, not counting a salary for the founder.
How much money is needed?	About $700 covers the build period, a safety buffer, the certificates for the desktop apps, and about four months after launch with no paying customers.
Where does it stand?	The core works on the computer: drawing, recording and exporting a video. Online features, AI features and billing are not built yet. There are no paying customers yet.
Biggest unknown?	How many free users will pay. The plan assumes 4%, but that figure is a guess and will be measured after launch.

1 · THE PROSPECT

Why there is an opening

Explainer videos are everywhere, but once one is recorded it is frozen. Flowcam's bet is that creators will pay for videos they can keep editing.

The problem, in plain words

A typo

In a normal screen recording, a spelling mistake on a diagram means recording the take again, or living with it.

A phone-shaped version

Turning a wide video into a tall one for phones (9:16) usually means cropping and losing part of the diagram.

The viewer

Viewers can watch the diagram, but they cannot open it, copy it or build on it.

This is the product's working idea. It has not yet been tested in interviews with users.

The gap in the market

The two closest products each lack one half of what Flowcam offers:

Product	Diagrams revealed step by step	Face camera in sync
ExcaliRec	No	Yes
Excalimate	Yes	No
Flowcam	**Yes**	**Yes**

Because each rival is "one feature away", speed matters: Flowcam needs to launch before they close the gap.

Why Flowcam can stay ahead: recipes, not photos

A normal recording is like a photo of a cake: you can look at it, but you cannot change the ingredients. Flowcam keeps the **recipe**. It records what was drawn and when each step appeared, then rebuilds the video from those instructions when you export. That makes three things possible that rivals cannot do just by adding a missing feature:

- **Fix after recording:** correct a typo or change a step's timing without recording again.
- **Re-export anything:** any resolution or colour theme, including a tall phone version that rearranges the diagram instead of cropping it.
- **Remix:** viewers of a shared video can open the diagram and copy it into their own account.

Why people would pay

The free version is useful on its own. People pay when they hit one of these "upgrade moments", each of which opens an upgrade panel in the app:

- They want to **share** a video by link, with viewers able to open the diagram.
- They want to **remove the watermark** or export in **4K**.
- They want **Screen Polish** in the desktop app (automatic zoom and smooth cursor).
- They run **out of Credits** for AI features.

2 · FEATURES

What users get

A user goes from an idea to a finished, shareable video in five steps. No account is needed to start; an account is needed only for AI features and Pro.

1. Start
Try a sample presentation, then pick a template, generate one from a topic with AI, or start blank.

2. Draw
Build hand-drawn diagrams across several pages, and set the order in which parts appear.

3. Present
Record face, voice and the diagram while clicking through the steps, with a script on screen.

4. Polish
Dead air is trimmed and captions are added automatically. Fine-tune on a timeline.

5. Export or share
Save an MP4 video file, or share a link where viewers can also open the diagram.

At launch (version 1, in the web browser)

Area	What it does
Drawing	Hand-drawn shapes, arrows (optionally animated), sticky notes, freehand, text and images. Handwritten fonts, including two for Bangla. 4–6 colour themes and about 10 templates for lessons, explainers and system design. Imports diagrams from Excalidraw.
Steps	Parts of a diagram appear one at a time. Steps are created automatically from the drawing order and can be reordered, merged or given effects such as drawing on, highlighting or zooming in.
Recording	Camera and microphone check before recording, a 3-2-1 countdown, a private presenter screen with the script and the next step, a teleprompter, keyboard shortcuts, temporary pen marks, several camera layouts, and retakes that pick up from the right step.
Editing	Trim, split, speed up and delete; transitions; backgrounds and frames; noise reduction and volume levelling; captions generated on the user's own computer for free, with subtitle file export.
Exporting	The MP4 file is made on the user's own computer. Free: watermarked, up to 1080p. Pro: no watermark, up to 4K.
Sharing (Pro)	A share link with a video player, "Open Board" and Remix, AI-written chapters, a transcript, embed code, an optional password and view counts.
Sync (Pro)	Work moves between devices. Conflicting edits are never overwritten silently.
AI (Credits)	Generate a whole presentation from a topic or a document, write scripts, and write titles, summaries and chapters for shared videos.

After launch

Release	Main additions
Version 1.1	Narrated videos: each step plays its own narration, recorded by the user or made by an AI voice. Script upload, AI images, transcript editing, basic screen recording in the browser, background music, title cards, and tall (9:16) and square (1:1) exports that rearrange the diagram.
Version 1.2	Desktop apps for Mac and Windows. Screen Polish for Pro (automatic zoom, smooth cursor, keyboard overlays, blurred areas), plus system audio, global shortcuts and a floating camera bubble. Short video clips inside diagrams (Credits).
Later	AI avatar, team workspaces, timestamped comments, Linux and mobile, among others.
Not planned	Several people editing the same diagram live, free multi-track video editing, and export back to Excalidraw.

2 · FEATURES

Free versus Pro

The rule is simple: everything that runs on the user's own computer is free; everything that needs Flowcam's online services is Pro.

	Free	Pro
Price	$0	$20 a month or $200 a year. Founding Price: $17 a month or $170 a year, kept for life.
Where work is kept	On the user's computer only	Online too: share links, sync across devices, 100 GB of storage
Video export	Watermark, up to 1080p	No watermark, up to 4K
Screen Polish	No	Yes, in the desktop apps
AI Credits	50, once, at sign-up	1,000 every month (also on yearly plans)
Trial	None	7 days, card required, 200 Credits

> Paying yearly costs $200 instead of $240 for twelve months: two months free. The Founding Price is for people who subscribe in the first 3 months after launch or among the first 500 subscribers, whichever comes first.

Money coming in

There are two sources of income: Pro subscriptions, which are the main business, and Credit top-ups for people who use a lot of AI.

Subscriptions

Pro costs $20 a month or $200 a year. To reward early customers, the first wave pays a **Founding Price** of $17 a month or $170 a year, and keeps that price for as long as they stay subscribed. All the plans in this report use the $17 Founding Price, which is the cautious choice.

Credits: tokens for AI tasks

AI tasks cost Flowcam real money from AI providers, so they are paid for with Credits. **One Credit is worth one cent**, and Credits are priced at about three times what each task costs Flowcam. Each task has a fixed price, with a "Best" quality option that costs more:

AI task	Standard	Best	What 1,000 monthly Pro Credits buy at Standard
Generate a whole presentation	5	15	200 presentations
Write a script	1	6	1,000 scripts
Match an uploaded script to steps	about 2	–	about 500 matches
Generate an image	12	45	83 images
AI voiceover, per minute	1	6	1,000 minutes
Video clip, per 5 seconds (1080p)	75	300	13 clips
Titles, summary and chapters	free	–	Included with share links

- **Top-ups:** 1,000 extra Credits for $10. They never expire.
- **Bonus Credits** (offers, referrals, apologies) expire after 90 days.
- **Order of use:** the monthly allowance first, then Bonus Credits, then top-ups.
- **Fair charging:** Credits are reserved when a task starts and charged only if it succeeds.

Taxes and payments

Payments go through **Lemon Squeezy**, which acts as the "merchant of record": it sells on Flowcam's behalf and collects and files sales tax and VAT in every country. Prices shown exclude tax; tax is added at checkout. Lemon Squeezy charges 5% plus 50 cents per payment, with small extra fees for international cards, subscriptions and payouts outside the US. Flowcam uses it because the business is based in Bangladesh, where Stripe does not accept businesses.

When someone cancels

Anyone who has paid at least once keeps working share links; their online items become view-only and downloadable. Share links made during a Trial that ends unpaid stop working, after an on-screen warning and an email two days before the Trial ends.

4 · WHAT IT COSTS

Money going out

Costs come in two kinds. **Fixed costs** are paid every month whatever happens. **Per-customer costs** grow with each paying customer.

Where each $17 goes



Costs: about $4.05 (24%)

Per paying customer, per month	Paid to	Amount
Payment processing and tax handling	Lemon Squeezy	about $1.85
Video streaming for share links	Mux	about $1.00
AI tasks (assuming about half the monthly Credits are used)	AI providers via OpenRouter	about $0.80
Database, accounts and file storage	Supabase	about $0.40
Total cost		**about $4.05**
Left from $17 ("contribution")		**about $13.00 (76%)**

At the $20 list price, about $15.70 (79%) is left. Payment fees range from about $1.60 to $1.85 depending on where the customer is.

The worst case: a customer who uses every Credit

If a Pro customer uses all 1,000 Credits every month, about **$10.40 (61%)** is still left. Because Credits are priced at about three times their cost, heavy AI users stay profitable.

Free users cost almost nothing

The heavy work (drawing, recording, making the video file and writing captions) runs on the user's own computer, not on Flowcam's servers. A free user's main cost is the 50 sign-up Credits: at most about $0.17, and about $0.04 on average because many are never used. A 7-day Trial costs about $0.30.

4 · WHAT IT COSTS

Fixed costs and the starting budget

Fixed monthly costs

Item	What it is for	Cost
Vercel Pro	Hosting the website and the app's online services	$20 / month
Supabase Pro	Database, user accounts, file storage	$25 / month
Business email	Email for the business	$7 / month
Total from launch		**$52 / month**
Apple Developer membership	Needed to sign the Mac app (from version 1.2)	$99 / year
Windows code-signing certificate	Needed to sign the Windows app (from version 1.2)	about $230 / year

> **Why the spec also says "about $90 a month".** The two desktop items add about $27 a month when spread over a year, and free users' AI Credits add a little more. The spec rounds this to about $90 a month and uses it for the break-even figure. These prices were quoted from memory and must be checked when the accounts are opened.

The starting budget: about $700

Phase	What is paid	Cost
Build, about 3 months before launch	Domain name (~$15 once), business email (~$7/month), AI testing (~$20/month). Other services stay on free plans.	about $100 total
Safety buffer	Prepaid AI credit, used up as needed	$50
Desktop launch (version 1.2)	Apple and Windows signing	about $329 / year
Running after launch	About $90 a month including free users' AI, with no paying customers	about four months

Together, about **$700** covers the build, the buffer, the desktop certificates and about four months after launch, even with zero paying customers. This excludes the founder's salary, computer and development tools.

Putting it together

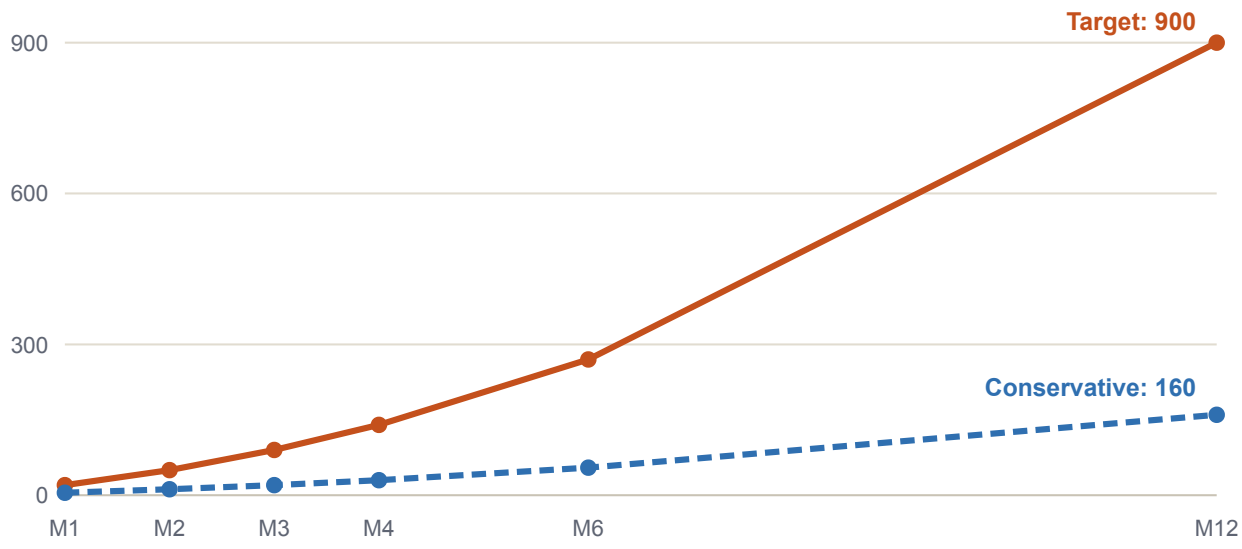
Break-even: about 7 customers

Each paying customer leaves about $13 a month. Fixed costs are about $90 a month. $90 ÷ $13 ≈ 7. So **about 7 paying customers cover the running costs**. Every customer after that adds about $13 a month of profit before salary.

Paying customers	Monthly profit before salary (at $17)
7	break-even
500	about $6,400
1,000	about $12,900

The first year: two scenarios Assumption

The plan models two paths. **Conservative** assumes 500 new free users and 25 Trials a month. **Target** assumes 1,000 new free users and 60 Trials a month. Both use the $17 Founding Price.



Paying customers by month after launch. Solid orange line: Target. Dashed blue line: Conservative. Points are the months given in the plan; the lines between them are straight joins.

Month after launch	Conservative: customers	Conservative: profit	Target: customers	Target: profit
1	5	-$15	20	+$149
2	12	+$76	50	+$538
3	20	+$180	90	+$1,056
4 (desktop certificates paid)	30	-$20	140	+$1,374
6	55	+$633	270	+$3,386
12	160	+$1,992	900	+$11,545
Whole first year		**about +$9,600**		**about +$55,600**

Read these numbers with care. Both scenarios assume that about 4% of free users become paying customers. No published figure for this exists, so 4% is a guess. It will be measured from launch and replaced after 60 days. Fixed costs stay at $52 a month in both scenarios, because neither reaches a point where bigger infrastructure is needed.

5 · ECONOMICS

What would change the numbers

Factor	Effect
Conversion rate	The biggest lever. If fewer than 4% of free users pay, more free users are needed to reach the same income. If more pay, the Target scenario comes sooner.
Price	Customers paying the $20 list price leave about $15.70 instead of $13.00. Once the Founding window closes, new customers pay $20.
AI usage	Heavy AI use lowers the margin from 76% towards 61%, but never below it, because the monthly Credit allowance is capped.
AI model prices	Credit prices are tied to the AI models' prices as of 17 October 2026. They must be reviewed whenever a model changes.
A video going viral	Without limits, one popular video could cost more than its owner pays. Section 6 explains the limit that prevents this.

6 · KEEPING COSTS UNDER CONTROL

Why costs stay low as Flowcam grows

The design keeps expensive work off Flowcam's servers and puts a ceiling on every cost that could run away.

1. Heavy work happens on the user's computer

Drawing, recording, making the video file and writing captions all run on the user's own device. Flowcam's servers mainly handle accounts, small files and AI tasks. This is why a free user costs almost nothing.

2. A cap on viral videos

Video streaming is charged per minute watched. One 5-minute video watched 100,000 times would cost Flowcam about $400, against the $17–20 its owner pays. So each Pro customer gets **10,000 watched minutes a month** on the premium video player. Beyond that, the link keeps working from a plain video file on Cloudflare R2, which charges nothing for downloads, and the owner gets a "your video is popular" email. Links not watched for 90 days also move to R2.

3. Spending limits and circuit breakers

Every paid provider has a spending cap, and Flowcam sets its own daily budgets. If spending runs away, a "circuit breaker" pauses that service automatically, like a fuse in a house.

4. Abuse protection for free Credits

- Sign-up Credits are given only after the email address is verified.
- A bot check runs on sign-up and on every AI task from a free account. Throwaway email addresses are blocked.
- Only one sign-up grant per device and network address in 24 hours.
- Limits per plan: Free 1 task at a time and 10 a day, with no video tasks on sign-up Credits; Trial 2 at a time and 30 an hour, with at most 3 video tasks; Pro 3 at a time and 60 an hour, with at most 20 video tasks a day.

5. Upgrade only when a trigger is reached

What could strain	Trigger	Planned upgrade
AI task processing	Waiting time over 30 seconds at peak	More workers, or a dedicated service
File storage and transfer	Over $300 a month	Move synced files to Cloudflare R2
Video streaming (Mux)	Over $1,000 a month	Volume pricing, or compare Cloudflare Stream
Website and online services (Vercel)	Over $500 a month	Move online services to Cloudflare Workers
Analytics	Over 1 million events a month	Record only a sample of less important events

7 · RISKS AND STATUS

What could go wrong, and where things stand

Risk	Why it matters	Current plan
Copycats	ExcaliRec or Excalimate could add the half they are missing.	Launch version 1 fast. The "recipe, not photo" design is harder to copy.
Conversion	Unknown; it decides how many free users are needed.	Measure sign-ups, Trials and upgrades from launch; review after 60 days.
Payments from Bangladesh	Stripe does not accept businesses based in Bangladesh.	Lemon Squeezy now. Consider a US company with Stripe once revenue justifies it.
Product name	"FlowCam" is already the name of a scientific instrument brand (Fluid Imaging Technologies, now Yokogawa).	Search the US and international trademark databases before buying the domain getflowcam.com.
Cost estimates	Some fixed prices were quoted from memory.	Check them when opening accounts; track actual invoices monthly.
Viral videos	Streaming costs could exceed what a customer pays.	The 10,000-minute monthly allowance, then plain files on R2. Still to be written into the terms of service.

Build progress

Done

- The hand-drawn drawing engine, adapted from an earlier project (Excaliflow)
- The editor for building presentations
- The recorder, which saves each recorded piece safely even if the browser crashes
- Exporting recordings to MP4 video files
- A pre-launch website with a waitlist

Not yet

- Online features: sync and share links
- AI tasks and Credits
- Billing
- Paying customers

Build progress is taken from the code history up to 17 October 2026. The recording work is on a development branch that has not yet been merged.

REFERENCE

Glossary

Term	Meaning
Presentation	A set of hand-drawn pages (Boards) that a user presents step by step.
Board	One page of a Presentation: a canvas with a hand-drawn diagram.
Step	One moment in a Presentation where part of the diagram appears or changes.
Clip	One continuous recorded piece. Pausing or retaking starts a new Clip.
Pro	The paid subscription: $20 a month, or $17 at the Founding Price.
Founding Price	$17 a month or $170 a year, kept for life by people who subscribe in the first 3 months or among the first 500.
Trial	7 days of Pro with 200 Credits; a card is required.
Credit	A token for AI tasks, worth one cent.
Share Link	A web link to a finished video. Viewers can also open the diagram and Remix it.
Remix	Copying a shared diagram into your own account to build on it.
Sync	Keeping work up to date across devices (Pro).
Screen Polish	Pro desktop features that make screen recordings look professional: automatic zoom, smooth cursor, keyboard overlays and blurred areas.
Viewing Allowance	10,000 watched minutes a month per Pro customer on the premium video player.
Contribution	What is left from a customer's payment after the costs that customer causes. It pays the fixed costs, then becomes profit.
Break-even	The point where income covers all running costs.
Merchant of record	A company that sells on your behalf and handles tax and payments. For Flowcam, this is Lemon Squeezy.
Vercel, Supabase, Mux, R2, OpenRouter	The outside services Flowcam rents: website hosting; database and storage; video streaming; low-cost file delivery; access to AI models.

Sources

docs/PRODUCT.md (agreed design, 17 October 2026), sections 1, 2, 5, 6, 7, 8 and 9 · docs/OPEN-ITEMS.md items 1–7 · decision records 0002, 0005, 0007, 0008, 0011, 0014, 0015, 0016 and 0017 · the git history of the Flowcam repository up to commit e13f3e5. AI and video prices were read from OpenRouter on 17 October 2026. The founder's private payment notes were not used.